

A/B Test Analysis Report

Boost Power Evaluation

Game Data Analysis
Delivery 2 - A/B Testing

Yiwei Ye

December 2025

1. Executive Summary

This report analyzes a 4-month A/B test conducted on a puzzle game to determine whether an increased boost power feature should be rolled out to all players.

Key Findings

- Players with stronger boost (Group A) achieved a 2.6 percentage point higher success rate (25.1% vs 22.5%)
- The difference is statistically significant ($p < 0.001$)
- Group A players used boosts 51% more frequently than Group B (15.6% vs 10.3%)
- The positive effect persisted even after the boost was reset to original power

Recommendation

Roll out the stronger boost to all players, with ongoing monitoring of long-term retention and monetization metrics.

2. Introduction & Background

2.1 Game Context

The game is a Candy Crush-style puzzle game where players attempt to complete levels. Players can use 'boosts' - power-ups that help them succeed at difficult levels. The game launched on January 1, 2020.

2.2 Business Problem

Players reported that some levels were too difficult, leading to frustration. The product team hypothesized that increasing the boost power would improve player experience and engagement without negatively impacting game balance.

2.3 Test Design

Group A (Test)	Received increased boost power
Group B (Control)	Retained original boost power
Test Period	June 1 - September 30, 2020
Sample Size	~14,000 players (6,923 A, 7,017 B)

2.4 Research Question

Should the increased boost power be rolled out to all players?

3. Methodology

3.1 Data Sources

Dataset	Records	Key Fields
Users	13,940	TestGroup, demographics
Sessions	432,000+	Session timing
Levels	1.5M+	Attempts, boost, outcomes
Transactions	26,000+	Purchase data

3.2 Time Period Segmentation

Period	Dates	Purpose
Pre-Test	Jan 1 - May 31, 2020	Baseline validation
During Test	June 1 - Sept 30, 2020	A/B comparison
Post-Test	Oct 1 - Dec 31, 2020	Persistence analysis

3.3 Statistical Methods

- Chi-squared test for comparing proportions between groups
- 95% Confidence Intervals for effect size estimation
- Odds Ratio for practical significance interpretation
- Significance threshold: $\alpha = 0.05$

4. Results & Analysis

4.1 Baseline Validation

Before analyzing test results, we verified that Groups A and B were comparable before the test began.

Metric	Group A	Group B	p-value
Sample Size	6,923	7,017	-
Pre-Test Success Rate	29.1%	30.1%	0.0016

Finding: There was a small but statistically significant difference in baseline success rates (1 percentage point). This is noted as a limitation, but the difference is small enough that it does not invalidate the test results.

4.2 Primary Finding: Level Success Rate

During the test period (June - September 2020):

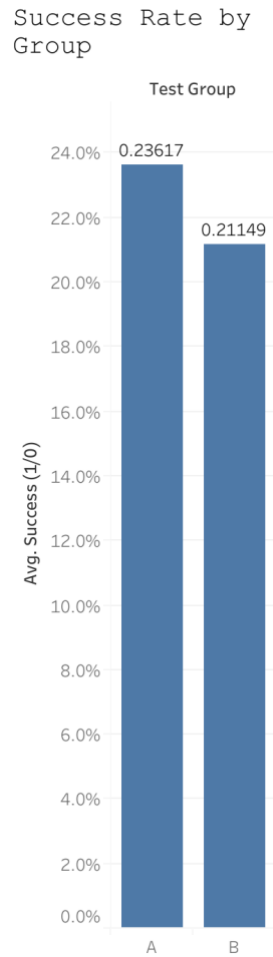


Figure 1: Success Rate by Test Group

Metric	Group A	Group B	Difference
Success Rate	25.1%	22.5%	+2.6 pp
Level Attempts	399,219	394,339	-

Statistical Test Results

- Chi-squared statistic: 797.84
- Degrees of freedom: 1
- p-value: < 2.2e-16 (highly significant)
- Odds Ratio: 1.15
- 95% CI for difference: [2.35%, 2.86%]

4.3 Time-Series Analysis

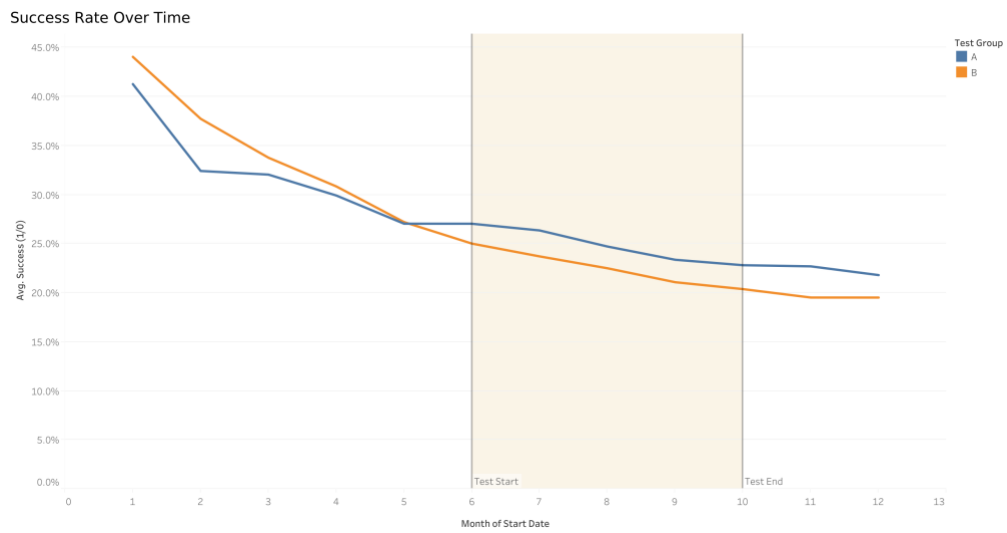


Figure 2: Success Rate Over Time by Test Group

The time-series visualization shows clear separation during the test period, with Group A consistently outperforming Group B. Notably, the effect persists even after the boost was reset in October.

4.4 Secondary Finding: Boost Engagement

Boost Usage Rate

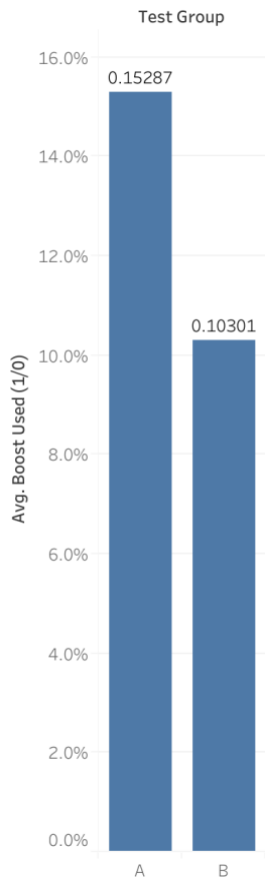


Figure 3: Boost Usage Rate by Test Group

Metric	Group A	Group B	Difference
Boost Usage Rate	15.6%	10.3%	+51%

Players used the boost significantly more when it was stronger ($p < 0.001$), indicating they recognized and appreciated the improved feature value.

4.5 Persistence Effect

Success Rate by Test Period

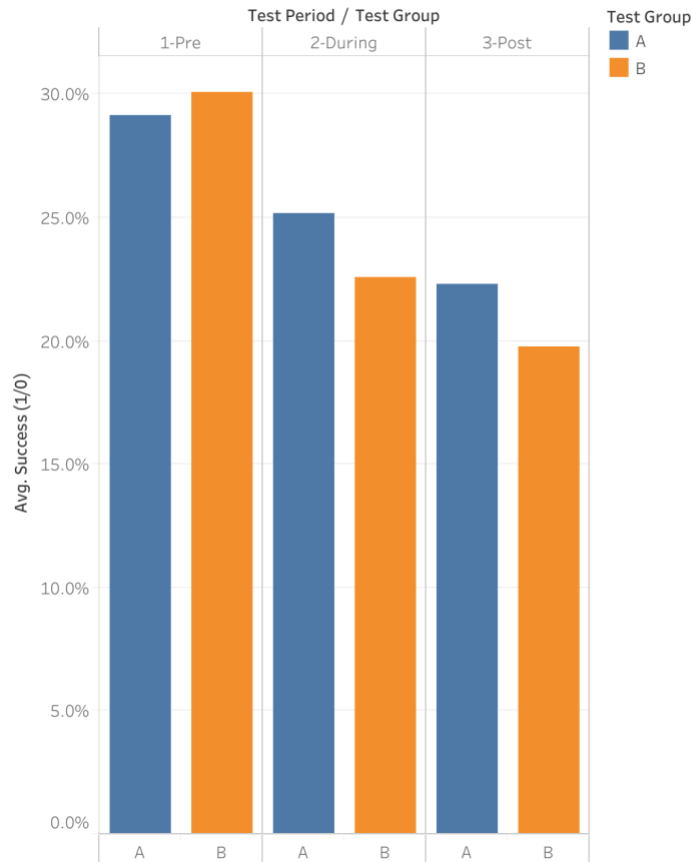


Figure 4: Success Rate by Test Period

Period	Group A	Group B	Difference
Pre-Test	29.1%	30.1%	-1.0 pp
During Test	25.1%	22.5%	+2.6 pp
Post-Test	22.3%	19.7%	+2.6 pp

Key Insight: Even after the boost was reset to original power, Group A continued to outperform Group B by the same margin. This suggests learned behavior or habit formation - players who experienced the stronger boost developed better strategies or confidence that persisted.

5. Discussion

5.1 Arguments FOR Rolling Out Stronger Boost

- Significant Performance Improvement: 2.6 percentage point increase with $p < 0.001$

- Higher Player Engagement: 51% increase in boost usage indicates feature value
- Lasting Positive Effects: Effect persists after reset, suggesting skill development
- No Observed Negative Effects: No decline in session frequency or engagement

5.2 Arguments AGAINST Rolling Out Stronger Boost

- May Reduce Game Challenge: Easier success could reduce long-term engagement
- Potential Boost Dependency: Players may become reliant on boosts
- Unknown Monetization Impact: Stronger free boost might reduce purchases
- Baseline Difference: Small pre-test difference suggests imperfect randomization

6. Conclusion & Recommendation

6.1 Summary of Findings

Test	Result	Significant?
Success Rate (During)	A > B by 2.6 pp	Yes (p < 0.001)
Boost Usage (During)	A > B by 51%	Yes (p < 0.001)
Baseline (Pre-Test)	Similar (1 pp diff)	Minor concern
Persistence (Post-Test)	Effect persists	Yes (p < 0.001)

6.2 Final Recommendation

RECOMMEND: Roll out the stronger boost to all players

The evidence strongly supports this decision:

- Clear, statistically significant improvement in player success
- Increased engagement with the boost feature
- Positive lasting effects on player behavior
- No observed negative consequences during the test period

6.3 Suggested Next Steps

- Monitor Long-Term Metrics: Track D7, D30, D90 retention after rollout
- Analyze Monetization Impact: Compare ARPU and boost purchase behavior
- Consider Level Rebalancing: Review difficulty curves if needed
- A/B Test Monetization: Test pricing changes if boost purchases decline